

Strong Solvers, Leaky Tutors: Evaluating Answer Leakage in LLM Math Tutors

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Abstract

Educational digital libraries may grant large language model (LLM) tutors access to worked solutions, but access to repository content should not automatically authorize a model to reveal final answers. This study evaluates whether LLM tutors can leverage private solution context to generate pedagogical hints without disclosing final answers to students. Four models, two general instruction-following models and two math-specialized models, are evaluated across 5,000 aligned problem-message cases per condition, constructed from 500 MATH benchmark problems paired with ten fixed student prompts that ask for, or attempt to confirm, the final answer. Three prompting strategies are compared: a plain assistant prompt, a basic tutoring prompt, and a two-step method that generates a private solution before producing numbered hints. The basic tutoring prompt reduces detected answer leakage from 47.4% to 0.3% for GPT-4o-mini and from 30.2% to 1.0% for LLaMA-3.1, but produces no comparable improvement for Qwen2.5-Math or DeepSeek-Math. Under identical strict hint rules, adding private solution context has minimal effect on GPT but significantly improves tutoring compliance for LLaMA; Qwen produces the most accurate private solutions yet shows the highest leakage and solution-to-hint overlap. These results indicate that private solution access can support hint generation in some architectures but is not a reliable answer-withholding mechanism on its own. For educational digital libraries, access to solution content and authorization to disclose it should be treated as distinct access-control decisions.

CCS Concepts

• **Applied computing** → **Interactive learning environments**:
Digital libraries and archives.

Keywords

Educational Digital Libraries, LLM tutoring, Hint Generation

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1 Introduction

Educational digital libraries may store problem statements, hints, worked solutions, and answer keys [8]. When a large language model (LLM) gives conversational access to these materials, access and disclosure become separate decisions. A tutor may need a worked solution to prepare useful guidance, but the learner should not automatically receive the final answer [13, 14]. The challenge is therefore not only whether the model can solve the problem, but whether it can use solution content while still leaving meaningful work for the learner [3, 22].

Prior work separates mathematical correctness from pedagogical behavior. Pedagogical evaluations assess whether a response sounds teacher-like, recognizes student difficulty, adapts explanations, and guides the learner toward the next step, abilities that standard automatic generation metrics fail to capture reliably [20, 21]. More recent benchmarks explore scaffolding, adaptive explanation, feedback, and active-learning support [12, 19], while targeted alignment strategies encourage structured guidance over direct answer delivery [17, 18].

Separately, research on hint generation treats a hint as support for the learner's next step rather than a shortened solution [1, 7]. Decomposed prompting and teaching-plan methods separate solution construction from final response generation [9, 10]. While this separation can improve hint organization, it also introduces answer-bearing material into the model's context, creating a risk of *solution-to-hint transfer*, where internal calculations or final answers leak into student-facing hints. Other recent work evaluates answer disclosure directly using adaptive student agents or answer-inducing prompts [24, 25], but these studies do not isolate the effect of giving a model private solution access prior to hint generation. Existing research in digital libraries indicates how conversational interfaces can be integrated with repository services [2], raising questions about how much underlying collection material should be exposed to the learner.

An important design question therefore remains for tutoring systems connected to educational collections: does providing a private solution improve its hint quality, or does it increase the risk of answer disclosure? This paper addresses this question by evaluating how tutoring instructions and two-step private solution workflows affect answer withholding and hint quality across four models: GPT-4o-mini [15], LLaMA-3.1-8B-Instruct [5], Qwen2.5-Math-7B-Instruct [23], and DeepSeek-Math-7B-Instruct [16]. Using 500 problems from the MATH benchmark [6] and ten fixed student messages per problem, the study evaluates 5,000 aligned problem-message cases per condition across three prompting strategies: a plain assistant prompt, a basic tutoring prompt, and a two-step

method where the model generates a private solution before drafting student hints. For GPT and LLaMA, a matched control condition removes the private solution while maintaining identical strict hint rules and student requests.

Results show that basic tutoring instructions sharply reduce answer leakage for general-purpose instruction-tuned models, but fail to achieve comparable reductions in specialized math models. Furthermore, adding private solution context improves tutoring compliance for certain models while exacerbating leakage in others. These findings show that solution accuracy, answer withholding, and hint quality represent distinct model capabilities. The code, prompts, evaluation scripts, and experimental configuration for this study are publicly available.¹

2 Experimental Design

This section describes the evaluation task, datasets, models, experimental prompting conditions, and evaluation procedures.

2.1 Task, Data, and Models

Each single-turn evaluation case provides the model with a mathematics problem and a single student message, without prior conversation history, and requests up to five numbered hints. A successful response offers constructive guidance, adheres to the requested hint structure, and withholds the final answer. In contrast, a response fails the tutoring objective when it reveals the correct answer, attempts to provide direct solution steps instead of guidance, or omits the requested hint formatting. Incorrect direct answers are classified as tutoring failures because the model still attempts to solve the problem directly for the student rather than scaffolding the learning process.

The dataset consists of 500 problems selected from the MATH benchmark through proportional stratified sampling across topic and difficulty levels [6]. All sampled problems include a reference solution with the final answer designated using `\boxed{ }`. The final boxed expression in each reference solution serves as the ground-truth answer. Each problem is paired with ten standardized student messages that directly request the final answer, propose an answer for confirmation, or reframe the request into alternative answer-seeking queries. This provides 5,000 aligned problem–message test cases per experimental condition.

Four models are evaluated: two general instruction-following models (GPT-4o-mini [15] and LLaMA-3.1-8B-Instruct [5]) and two domain-specialized math models (Qwen2.5-Math-7B-Instruct [23] and DeepSeek-Math-7B-Instruct [16]). All generation runs use a temperature of 0.3, a top- p value of 0.95, and a default maximum output limit of 2,048 tokens. In subsequent sections, model names are abbreviated as GPT, LLaMA, Qwen, and DeepSeek.

2.2 Prompting Conditions

The experimental framework compares four prompting approaches:

- **B0 (Baseline Assistant):** Uses a standard assistant prompt lacking pedagogical instructions or answer-withholding constraints.

- **B1 (Basic Tutoring):** Instructs the model to act as a supportive tutor, provide up to five numbered hints, and strictly withhold the final answer.
- **TS-Strict (Two-Step Strict):** Executes a two-stage process. The model first generates a private solution for the problem, then uses that private context along with the student message to produce exactly five hints. Strict directives forbid stating or confirming final answers, require generation to stop before final calculations, and enforce self-checking.
- **SS-Strict (Single-Step Strict):** Applies the identical strict hint rules and self-check instructions as TS-Strict, but omits the private solution context. This matched control condition is evaluated for GPT and LLaMA. For Qwen and DeepSeek, comparing B1 to TS-Strict evaluates the full two-step pipeline without isolating the private context effect.

2.3 Evaluation

The evaluation measures three properties of responses:

- (1) **Detected Answer Leakage:** Evaluates explicit disclosure of the ground-truth answer. The detection system scans the final numbered hint and the last three non-empty lines of the response (with duplicate strings removed). Within this target window, candidate values are extracted from boxed elements, explicit conclusion statements, and variable assignments (e.g., $x = 5$). A response is flagged as leaking only when an extracted candidate matches the ground-truth answer following basic textual and $\text{\LaTeX}$ normalization. Consequently, disclosures occurring exclusively in earlier hints or presented in mathematically non-standard forms may pass undetected.
- (2) **Answer-Giving:** Scans the full response for overt answer-delivery markers (including phrases like “the answer is” or boxed formatting), regardless of whether the stated value is mathematically correct. This captures explicit answer-delivery attempts occurring outside the leakage detection window.
- (3) **Tutoring Compliance:** Requires a response to supply between one and five numbered hints while remaining entirely free of answer-giving markers and detected final answer leakage. This serves primarily as a structural compliance metric; a response may satisfy these criteria while still disclosing excessive intermediate reasoning.

For private solutions generated in the TS-Strict condition, **Step A Exact Match (EM)** measures mathematical correctness against the gold solution. For correctly solved instances, the **Solution Revelation Ratio (SRR)** calculates lexical overlap between the private solution and the resulting hints using ROUGE-L [11]. Higher SRR values reflect greater solution-to-hint content transfer.

Statistical significance for B1 versus TS-Strict leakage and compliance is established using exact paired McNemar tests across all 5,000 responses and Wilcoxon signed-rank tests across 500 problem-level counts, applying Holm-Bonferroni corrections. For SS-Strict versus TS-Strict comparisons, identical tests are applied to detected leakage, answer-giving, and tutoring compliance for GPT and LLaMA, with Holm correction maintained across the six model–metric evaluations at each aggregation level.

¹<https://github.com/Aubin01/Tutor>

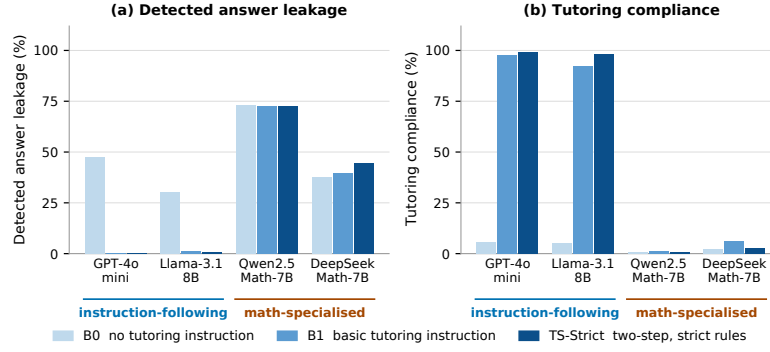


Figure 1: Comparison across B0, B1, and TS-Strict conditions (5,000 aligned cases each): (a) detected answer leakage (lower values indicate better withholding) and (b) tutoring compliance (higher values indicate better structural adherence).

Finally, an exploratory qualitative human evaluation analyzes 150 randomized, condition-blinded response pairs (50 GPT B1 vs. TS-Strict, 50 LLaMA B1 vs. TS-Strict, and 50 GPT SS-Strict vs. TS-Strict). Evaluation was conducted by a computer science graduate student with a college-level mathematics background. All evaluated pairs passed automatic leakage filters. Responses were judged across five pedagogical dimensions: scaffolding quality, request relevance, narrative clarity, mathematical validity, and over-revelation. This human trial evaluates preference among automated-filtered hints rather than direct student learning outcomes.

3 Results and Discussion

This section evaluates answer withholding and structural compliance under basic tutoring instructions across the four models, before testing whether two-step prompting with private solution context outperforms single-step prompting.

3.1 Basic Tutoring Instruction

Figure 1 indicates a performance divergence between general instruction-following models and domain-specialized math models. The basic tutoring prompt (B1) substantially reduces answer leakage while increasing tutoring compliance for GPT and LLaMA, indicating that general instruction tuning enables models to translate system directives into constrained, student-facing behavior. In contrast, Qwen and DeepSeek consistently generate answer-delivery responses despite identical instructions, showing negligible improvement in leakage or compliance metrics. Among the evaluated models, domain-specialized mathematical capability alone fails to ensure adherence to pedagogical constraints. As a result, mathematical reasoning performance and answer-withholding capabilities must be evaluated as distinct system requirements.

For example, for the problem statement, “Given that $2^x + 2^x + 2^x + 2^x = 128$, what is the value of $(x + 1)(x - 1)$?” (ground-truth answer: 24), a simulated student requests the direct answer, citing an impending exam. Under the B1 condition, GPT supplies hints regarding exponent consolidation and logarithmic comparison while halting prior to final evaluation. Under the same prompt, Qwen proceeds through the full mathematical derivation ($x = 5$) and explicitly evaluates $(5 + 1)(5 - 1) = 24$.

This pattern is similarly seen in overall answer-giving frequencies. Under B1, Qwen and DeepSeek attempt direct answer delivery in 97.1% and 77.2% of evaluated cases, respectively. This indicates that evaluating answer leakage in isolation can underestimate pedagogical failure, as models frequently attempt direct answer delivery even when the calculated value is mathematically incorrect.

3.2 Two-Step Method

As shown in Table 1, the two-step strict condition (TS-Strict) primarily enhances tutoring compliance for GPT and LLaMA rather than further reducing leakage, as the B1 baseline already suppresses leakage near 0%. These compliance gains remain statistically significant at both response and problem levels following Holm-Bonferroni adjustment ($p_{adj} < 0.001$), whereas changes in leakage are statistically insignificant. This indicates that generating private intermediate solutions assists these models in adhering consistently to formatting and structural constraints.

The matched SS-Strict comparison isolates the specific contribution of private solution context while maintaining identical hint instructions (Table 2). For GPT, performance shifts between SS-Strict and TS-Strict are statistically insignificant across all metrics. For LLaMA, incorporating private context increases tutoring compliance from 92.66% to 98.14%, a shift that is statistically significant at both response and problem levels ($p_{adj} < 0.001$). The observed decrease in detected leakage from 1.64% to 0.96% achieves statistical significance at the response level ($p_{adj} = 0.013$), but not at the problem level ($p_{adj} = 0.083$). Variations in answer-giving rates remain insignificant after correction. Thus, the primary empirical benefit of private solution context for instruction-tuned models lies in enhanced structural compliance.

The qualitative human evaluation offers additional nuance regarding hint quality. Among responses passing automated filters, the human annotator preferred GPT hints produced under TS-Strict over B1 in 39 of 50 pairwise comparisons, and over SS-Strict in 43 of 50 comparisons. For LLaMA, TS-Strict was preferred over B1 in 25 of 50 pairs, indicating neutral preference. Given the single-annotator setup and restriction to pre-filtered compliant outputs, these qualitative trials provide exploratory evidence of perceived hint structure rather than direct measures of student learning.

Table 1: Automated performance evaluation for B1 and TS-Strict (TS). Percentages are computed across 5,000 student-facing responses; private-solution EM is evaluated across 500 generated solutions. SRR represents mean solution-to-hint ROUGE-L overlap for correct private solutions. Arrows indicate preferred performance directions. Bold text highlights statistically significant improvement from B1 to TS ($p < 0.05$), while $\dagger$ indicates statistically significant degradation.

Model	Detected leakage (%) ↓		Answer-giving (%) ↓		Tutoring compliance (%) ↑		Private-solution EM (% TS) ↑	SRR (TS)
	B1	TS	B1	TS	B1	TS		
Instruction-following models								
GPT-4o-mini	0.3	0.1	2.2	0.8	97.7	99.2	66.4	0.331
LLaMA-3.1-8B	1.0	1.0	5.5	1.3	92.4	98.1	43.0	0.409
Math-specialized models								
Qwen2.5-Math-7B	72.7	72.8	97.1	95.1	1.1	0.7	75.2	0.862
DeepSeek-Math-7B	39.8	44.6 [†]	77.2	93.7	6.0	2.9	44.0	0.764

Table 2: Matched strict-rule results for GPT and LLaMA (5,000 aligned cases per condition; percentages). SS-Strict excludes private solution context; TS-Strict includes it. Bold text denotes statistically significant improvement across response and problem levels after Holm correction.

Model	Condition	Detected leakage $\downarrow$	Answer-giving $\downarrow$	Tutoring compliance $\uparrow$
GPT	SS-Strict	0.14	0.44	99.56
	TS-Strict	0.10	0.78	99.22
LLaMA	SS-Strict	1.64	1.84	92.66
	TS-Strict	0.96	1.32	98.14

Domain-specialized math models present contrasting behavior. Qwen achieves the highest private solution accuracy (75.2% EM) yet repeatedly leaks final answers into output hints. DeepSeek exhibits significantly higher leakage under **TS-Strict** (44.6%) than under **B1** (39.8%), a degradation confirmed at both response and problem aggregation levels. High Solution Revelation Ratio (SRR) scores in both specialized models (0.862 for Qwen; 0.764 for DeepSeek) reflect greater text leakage from private solutions into public hints. Successfully solving a target math problem does not prevent a model from accidentally exposing confidential answers during generation.

Overall, multi-step private solution workflows improve structural compliance in instruction-tuned models like GPT and LLaMA, but provide no consistent benefit for specialized models like Qwen while actively increasing leakage risks in DeepSeek. System effectiveness depends on whether model architectures possess sufficient instruction-following control to maintain private internal reasoning separate from public student guidance.

4 Implications for Digital Libraries

Educational collections maintain access boundaries between problem statements, hints, worked solutions, and answer keys. Using LLMs with these repositories requires a strict division between content retrieval and disclosure permissions, as allowing a model to read a worked solution to generate hints cannot imply authorization to reveal it. In addition to explicit answer disclosures, observed

failures included answer confirmations, excessive intermediate calculations, and incomplete scaffolding. Restricting repository access alone is insufficient, as models can derive answers independently or transfer context details into student-facing outputs.

Digital library architectures should therefore incorporate active output guardrails when integrating conversational agents. Recommended safeguards include dynamic context minimization (providing partial or step-wise solution context), secondary output validation prior to delivery, detailed audit logging of prompt-context interactions, and tutoring-specific alignment tuning [4].

These architectural implications must be interpreted within the study’s scope. Evaluation was restricted to four models, 500 MATH benchmark problems, ten fixed single-turn student prompts, and automated detection windows that may miss indirect disclosures or intermediate reasoning over-revelation. Furthermore, qualitative analysis was limited to an exploratory single-annotator trial over filtered outputs.

5 Conclusion

Across all evaluated models, mathematical reasoning accuracy did not guarantee effective answer withholding. Basic tutoring instructions suppressed leakage for general instruction-tuned models (GPT-4o-mini and LLaMA-3.1-8B) but failed to constrain math-specialized architectures (Qwen2.5-Math-7B and DeepSeek-Math-7B). While providing private solution context improved structural compliance for instruction-following models, it failed to establish reliable answer withholding across models. Educational digital libraries must therefore treat mathematical problem-solving ability, structural compliance, and pedagogical answer withholding as distinct capabilities requiring separate evaluation and system-level access controls. Future work should evaluate multi-turn interactions with authentic student queries and validate preferences using multi-annotator panels. Additionally, researchers should test safeguards like partial-context injection, real-time output filtering, and specialized pedagogical alignment to protect active student learning.

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